

Nullish Coalescing Operator (??) - Simple Hinglish Summary

Source: <https://javascript.info/nullish-coalescing-operator>

Main baat

(??) (do question marks) ek **naya operator** hai (purane browsers me polyfill chahiye). Ye **pehli "defined" value** chunne ka short tarika hai.

"Defined" ka matlab: jo value **na** (null) **ho**, **na** (undefined).

Kaise kaam karta hai

(a ?? b) ka result:

- (a) **defined** hai to → (a)
- (a) **defined** nahi hai ((null)/(undefined)) to → (b)

Ye is code ka short form hai:

```
javascript
```

```
result = (a !== null && a !== undefined) ? a : b;
```

Sabse common use: Default value dena

```
javascript
```

```
let user;  
alert(user ?? "Anonymous"); // Anonymous (user undefined hai)  
  
let user2 = "John";  
alert(user2 ?? "Anonymous"); // John
```

Kai (??) **chain** karke list me se pehli defined value le sakte ho:

```
javascript
```

```
let firstName = null;  
let lastName = null;  
let nickName = "Supercoder";  
  
alert(firstName ?? lastName ?? nickName ?? "Anonymous"); // Supercoder
```

`||` se comparison (sabse important fark)

- `||` → pehli **truthy** value return karta hai
- `??` → pehli **defined** value (na null, na undefined) return karta hai

`||` `false`, `0`, `""`, `null`, `undefined` sabko **ek jaisa (falsy)** maanta hai. Lekin kabhi kabhi hume default value **sirf tab** chahiye jab value `null/undefined` ho, yaani jab value sach me set na ho.

javascript

```
let height = 0;
```

```
alert(height || 100); // 100 (0 falsy hai, isliye galat default aaya)
```

```
alert(height ?? 100); // 0 (0 defined hai, isliye sahi)
```

`0` height kai baar **valid value** hoti hai, use default se replace nahi karna chahiye. Isliye yahan `??` sahi kaam karta hai.

Precedence (kaun pehle chalega)

`??` ki precedence `||` jaisi hi hai (dono 3). Yaani `+`, `*` jaise operators pehle chalte hain, `=` aur `?` baad me.

Isliye expressions me **brackets lagana zaruri** hai:

javascript

```
let height = null;
```

```
let width = null;
```

```
let area = (height ?? 100) * (width ?? 50);
```

```
alert(area); // 5000
```

Brackets ke bina `*` pehle chal jayega aur galat result aayega:

javascript

```
let area = height ?? 100 * width ?? 50;
```

```
// ye aise chalta hai (galat): height ?? (100 * width) ?? 50
```

`&&` ya `||` ke saath `??` (forbidden)

Safety ke liye JS **bina brackets ke** `??` ko `&&` ya `||` ke saath use karne nahi deta:

javascript

```
let x = 1 && 2 ?? 3; // Syntax error
```

Brackets lagao to chalega:

javascript

```
let x = (1 && 2) ?? 3;  
alert(x); // 2
```

Summary

- `??` pehli defined (na null, na undefined) value chunne ka chhota tarika hai.
- Default values dene ke liye use hota hai:

javascript

```
height = height ?? 100; // height null/undefined ho to 100
```

- Precedence kam hai, isliye expressions me **brackets** lagao.
- `||` ya `&&` ke saath **bina brackets ke use nahi kar sakte**.

Quick Comparison

Operator	Kya check karta hai	<code>0</code> ke saath	<code>""</code> ke saath
<code> </code>	Truthy / falsy	<code>0</code> ko skip karta hai	<code>""</code> ko skip karta hai
<code>??</code>	<code>null</code> / <code>undefined</code>	<code>0</code> ko rakhta hai	<code>""</code> ko rakhta hai

Yaad rakho: `0` ya `""` valid value ho sakti hai to `||` ki jagah `??` use karo.